

Sikun Xu

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RESEARCH INTERESTS	Machine Learning and AI Systems; Causal Inference; Data-Driven Decision-Making; Recommendation Systems; LLM and Generative Models	
EDUCATION	Washington University in St. Louis • Ph.D. Candidate, Olin Business School • Advisors: Raphael Thomadsen and Dennis Zhang	2021–Present
	Columbia University in the City of New York • M.S. in Operations Research	2019–2020
	Shanghai Jiao Tong University • B.S. in Industrial Engineering	2015–2019
JOB MARKET PAPERS	• The Winner’s Curse in Data-Driven Decision Making: Evidence and Solutions. Sikun Xu, Raphael Thomadsen, and Dennis Zhang. <i>Submitted</i>. • SOTA or Luck? The Winner’s Curse in LLM Leaderboards. Sikun Xu, Raphael Thomadsen, Dennis Zhang, and Heng Zhang. <i>Submitted to NeurIPS 2026</i>.	
WORKING PAPERS	1. A Causal Approach to Representation Learning for Unstructured Data. Sikun Xu , Zhenling Jiang, Zhengling Qi, and Dennis Zhang. <i>Major revision at Management Science</i> .	
CONFERENCE PROCEEDINGS	1. Xu, Sikun , Ruoyi Ma, Daniel K. Molzahn, Hassan Hijazi, and Cédric Jozs. “ Verifying global optimality of candidate solutions to polynomial optimization problems using a determinant relaxation hierarchy. ” In 2021 60th IEEE Conference on Decision and Control (CDC), pp. 3143-3148. IEEE, 2021.	
WORK IN PROGRESS	1. Optimal Real-Data Allocation for Synthetic-Data-Augmented Inference, with Zineng Xu and Mingduo Zhao, <i>Submitted to NeurIPS 2026</i>. 2. Exploration Without Noise: Quality-Gated Learning in Platform Recommendations, with industry partners. 3. Generative Learning-to-Rank in Recommendation Systems, with industry partners.	
INDUSTRY EXPERIENCE	• TikTok Inc., San Jose, Machine Learning Scientist Intern • Sauma Capital LLC, New York, Quantitative Researcher • Capstone Investment Advisors, New York, Investment Intern • Cardinal Operations, Shanghai, Algorithm Engineer Intern	2026 2021 2020 2018

INVITED SEMINARS	• Washington University in St. Louis, Department of Statistics and Data Science 2026 • The Chinese University of Hong Kong, Business School 2025
CONFERENCE PRESENTATIONS	SOTA or Luck? The Winner's Curse in LLM Leaderboards
	• Conference on Information Systems and Technology, CA (Scheduled) 2026
	The Winner's Curse in Data-Driven Decision Making: Evidence and Solutions
	• INFORMS Annual Meeting, San Francisco, CA (Scheduled) 2026 • INFORMS Annual Meeting, Atlanta, GA 2025 • ISMS Marketing Science Conference, Washington, D.C. 2025 • Conference on Artificial Intelligence, Machine Learning, and Business Analytics, New Haven, CT 2024
	A Causal Approach to Representation Learning for Unstructured Data
	• ISMS Marketing Science Conference, Washington, D.C. 2025 • INFORMS Annual Meeting, Seattle, WA 2024 • POMS Annual Conference, Minneapolis, MN 2024 • INFORMS Annual Meeting, Phoenix, AZ 2023
	Data-Driven Security Selection for Wealth Management
	• POMS Annual Conference, Orlando, FL 2023 • INFORMS Annual Meeting, Indianapolis, IN 2022
	Verifying Global Optimality of Candidate Solutions to Polynomial Optimization Problems Using a Determinant Relaxation Hierarchy
	• INFORMS Annual Meeting, Anaheim, CA 2021
TEACHING EXPERIENCE	Guest Lecturer
	• MGT 680E AI & Machine Learning Business Applications, Washington University in St. Louis Fall 2024 • IEOR 4721 AI Application in Finance, Columbia University Spring 2022 • IEOR 4721 AI Application in Finance, Columbia University Summer 2021
	Teaching Assistant
	• Olin Business School, Washington University in St. Louis – MGT 680E AI & Machine Learning Business Applications – SCOT 519E Revenue Management – SCOT 500M Supply Chain Analytics: Stochastic Models – SCOT 558 Advanced Operations Strategy – SCOT 5704 Operations Management – SCOT 356 Operations and Manufacturing Management – SCOT 400D Supply Chain Management – SCOT 500D Project Management • Columbia University in the City of New York

- IEOR 4742 Deep Learning
- IEOR 4525 Machine Learning

ACADEMIC SERVICE

- Reviewer: *Journal of Investment Strategies, Conference on Information Systems and Technology (CIST)*
- Session Chair: INFORMS Annual Meeting

REFERENCES

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The Winner’s Curse in Data-Driven Decision Making: Evidence and Solutions

Data-driven decision making involves estimating the value of each potential option and selecting the one with the highest estimated efficacy. This approach underpins a wide array of modern marketing and AI applications. However, several papers have shown that the estimated effectiveness of the chosen options will be systematically over-optimistic, even when the estimated outcomes are themselves unbiased. Using simulations calibrated to realistic parameters from marketing papers, we first demonstrate that the magnitude of the winner’s curse is often high in real-life contexts, and that the severity of the winner’s curse depends on the true performance difference between options relative to the noise level in the data, the number of candidate options, and the number of observations. We further show that using machine learning to target treatments to different consumers can lead to extremely high levels of winner’s curse, especially if the machine learning model is very flexible. We propose a correction method based on an m -out-of- n bootstrap, and demonstrate that our bootstrap usually outperforms the solutions that have been previously proposed in the literature. Finally, using over 32,000 real-life A/B tests from the Upworthy Research Archive, we document substantial winner’s curse that accounts for over 70% of the estimated lift.

SOTA or Luck? The Winner’s Curse in LLM Leaderboards

Public LLM leaderboards are now central to how models are evaluated, compared, and publicized, yet we show that the reported performance of top-ranked models can be systematically overstated. The reason is a leaderboard winner’s curse (WC): benchmark scores are noisy averages over finite task sets, so ranking models in the decreasing order of scores rewards sampling luck alongside genuine ability. This overstatement is large enough to make the apparent winner statistically fragile: across four out of the five SWE-bench variants we study, the published rank 1 model has less than a 50% bootstrap probability of remaining rank 1 under task resampling. We develop an m -out-of- N task bootstrap for iid benchmark tasks while allowing arbitrary within-task correlation among models. We prove that the worst-case rank-induced bias (WC as a special case for the first rank) is $\Theta(1/\sqrt{N})$ in the number of tasks N , and that our correction reduces it to $o(1/\sqrt{N})$. We also introduce a rank probability matrix that replaces a single deterministic ranking with a distribution over plausible rankings. Empirically, rank-1 inflation reaches 1–3 percentage points across eight benchmark settings spanning code generation and agentic customer service. On SWE-bench Verified, selection bias explains 47% of the winner’s apparent top-five lead, and the top 28 models are statistically indistinguishable at the 5% level. Our results suggest that benchmark designers should either enlarge the task pools substantially to suppress the WC or report bias-corrected scores and ranking uncertainty alongside raw rankings.

A Causal Approach to Representation Learning for Unstructured Data

The increasing availability of unstructured data (e.g., images) in business and economics research has created new opportunities to control for confounders. A common approach is embedding-then-inference, where unstructured data is compressed into low-dimensional embeddings and incorporated into causal models. However, we show that this method can introduce significant bias because representation learning models optimized for reconstruction may miss relevant confounders. To address this, we propose causal embedding (CE), which explicitly aligns the objective of representation learning with the causal task by jointly predicting both treatment and outcome variables. This approach captures confounding information while maintaining low-dimensional efficiency and accommodates various embedding methods, including fine-tuned pretrained models. Simulations demonstrate that CE estimators outperform both embedding-then-inference and direct adjustment with double machine learning (DML) in subsequent causal inference tasks. We further demonstrate the practical value of CE in an online marketplace application, where we fine-tune a multimodal architecture combining a pretrained vision model, DINOv2, with a pretrained language model, Qwen3.

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